



Vidyavardhini's College of Engineering and Technology
Department of Computer Engineering

Mini-project presentation on **PREDICTING THE SPORT MARKET BUSINESS USING ML**

Industry Project: BinaryStich Software Vasai

TEAM MEMBERS

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PROBLEM STATEMENT

The sports industry faces major challenges in accurately predicting market dynamics such as ticket sales, merchandise demand, sponsorship ROI, and broadcast viewership. Traditional methods rely on intuition and static reports rather than data-driven insights.

- This leads to:
- Unpredictable revenue fluctuations
- Inefficient inventory and resource allocation
- Missed marketing and sponsorship opportunities

Machine Learning can address these challenges by uncovering hidden patterns in large datasets to make reliable and real-time predictions.

OBJECTIVES

- To develop predictive models for forecasting sports market trends.
- To analyze the impact of team performance, fan sentiment, and marketing efforts on business outcomes.
- To optimize ticket pricing, merchandise planning, and sponsorship strategies using ML predictions.
- To design a dashboard or automated system for decision-makers in the sports industry.

PROPOSED SOLUTION

- Collect and analyze multi-source data including match statistics, ticket sales, fan engagement, and social media sentiment.
- Apply Machine Learning algorithms such as Random Forest, XGBoost, and LSTM for trend forecasting.
- Build a predictive model that outputs ticket demand, merchandise sales, and ROI estimates.
- Visualize predictions through an interactive dashboard (Streamlit/Flask) for stakeholders.

Outcome: Data-driven decision support system for sports business forecasting.

IMPLEMENTATION PROCESS



TECHNOLOGIES USED

Category	Technologies / Tools
Programming Language	Python
Libraries	Pandas, NumPy, Scikit-learn, XGBoost, TensorFlow / Keras, etc
Data Visualization	Matplotlib, Seaborn, Plotly
Deployment	Streamlit / Flask
Data Sources	Sports APIs, Social Media(API), Ticketing Databases

SCOPE OF PROJECT

- Applicable to sports leagues, teams, merchandise retailers, and broadcasters.
- Scalable for multiple sports (cricket, football, basketball, etc.).
- Future expansion to include real-time analytics and reinforcement learning models.
- Can support strategic decision-making for marketing, event management, and sponsorship evaluation.